

Tarea 1

OPT1: Optimizacion

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Problem 1**24.3.7**

Let $\alpha = \sqrt{2 + \sqrt{2}} \in \mathbb{C}$.

a) Compute $f = \min_{\mathbb{Q}}(\alpha)$.

b) Find $E \subseteq \mathbb{C}$ such that E is the splitting field for f over $\mathbb{Q}$. Compute $|E : \mathbb{Q}|$.

c) Show that $\text{Gal}(E/\mathbb{Q})$ contains an element of order 4.

a) Consider $\alpha^2 - \sqrt{2} = 2$. Rearranging, we obtain $\alpha^2 = 2 + \sqrt{2}$. We thus have $\alpha^4 = (2 + \sqrt{2})^2 \Rightarrow \alpha^4 - 4\sqrt{2} - 6 = 0$. We can rewrite $4\sqrt{2} + 6$ in terms of α^2 , obtaining $4\sqrt{2} + 6 = 4\alpha^2 - 2$.

Therefore $\alpha^4 - 4\alpha^2 + 2 = 0$. As such we obtain a potential minimal polynomial $x^4 - 4x^2 + 2$. Using Eisenstein's, we see it's irreducible and monic with α as a root. Therefore $f = x^4 - 4x^2 + 2$.

b) We can find the roots as $\pm\sqrt{2 + \sqrt{2}}, \pm\sqrt{2 - \sqrt{2}}$. We note that because they share the same minimal polynomial, $|E : \mathbb{Q}| = \deg(\min_{\mathbb{Q}}(\alpha)) = 4$, and we have that $E = \mathbb{Q}(\sqrt{2 + \sqrt{2}}, \sqrt{2 - \sqrt{2}})$.

c) Consider the automorphism $\sigma \in \text{Gal}(E/\mathbb{Q})$ such that $\varphi(\alpha) = \beta$, where $\beta = \sqrt{2 - \sqrt{2}}$. Consider

$$\varphi(\sqrt{2}) = \varphi(\sqrt{\alpha\beta}) = \varphi(\alpha^2 - 2) = \varphi(\alpha)^2 - \varphi(2) = \beta^2 - 2 = -\sqrt{2}$$

Using this, we see that we can use σ recursively to obtain $\beta, \alpha, -\alpha, -\beta$. We thus show that σ is an element of order 4 in $\text{Gal}(E/\mathbb{Q})$. ♥